MP FSM

Donnerstag, 15. Oktober 2026 16:40

***Q2.1** Finite State Machines - Filament replacement (2 pts.)*

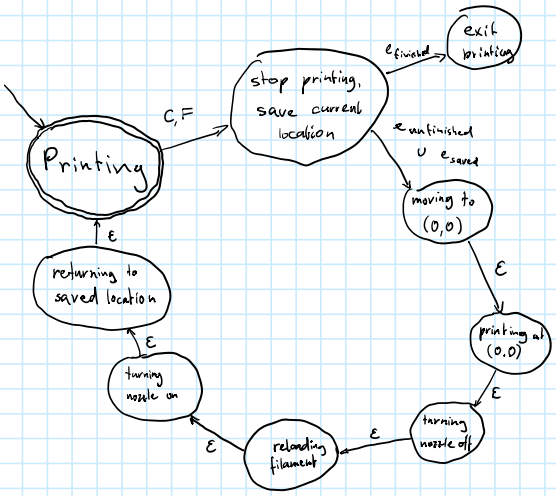
Next, you will model some basic high-level control operations of your 3D printer. Namely, you will model the replacement of filament in these two situations:

- When a new colour is required to be printed (C), or
- When the currently being used spool is finished (F).

We start modelling these two distinct situations as being triggered by two, independent, discrete signals C and F. Once either of these events is detected, the following set of steps needs to be taken:

1. Printing stops, saving the current location of the print head.
2. Moving to a wipe-tower location (x_p, y_p), which we assume to be at $(0, 0)$.
3. Excess filament is wiped (by printing on the wiping location).
4. The nozzle heat is switched off.
5. New filament is loaded.
6. The nozzle heat is switched on.
7. Printing is resumed.

Model the operation first as a Finite State Machine (FSM) of the 3D printer, including a state for "printing" and any necessary states to cover the filament replacement process.



***Q2.2** Automata - Parked detection (3 pts.)*

Your colleague forgot to order end-of-line sensors, but thankfully, the motors include rotational sensors you can use as a virtual end-of-line sensor. Simply put, you can detect an end-of-line by testing whether or not the motors are rotating to produce a vertical or horizontal motion when the corresponding voltages are applied to them. Let us assume the printer head is always brought to the origin via the concatenation of two types of motions:

- Vertical (V) until we hit the bottom of the printing area,
- Horizontal (H) until we hit the left-most side of the printing area.

Furthermore, consider that your design includes a system that periodically checks the value of the voltages applied (to determine the type of motion commanded), and the angular velocity sensors of the motors.

Consider signal sequences $\sigma : \mathbb{N} \to M \times m$ where $M = \{H, V\}$, and $m = \{+, 0\}$. A pair (M_i, m_i) denotes "type of motion" (horizontal, vertical), and "angular velocity" (positive or zero). As an example, one such sequence could be $\sigma = (H,+)(V,+)(H,+)(H,0)(V,+)(V,0)$.

Construct an automaton that could be used as a *head-parked detector* by accepting the sequences σ that lead to the printer head being correctly parked. Note that the same automaton with minor adjustments can be used to detect any of the other three corners of the printing area.

